RVI to NDVI Model Report

Synthetic Optical Vegetation Index Generation using Radar Polarizations

This technical report provides a detailed breakdown of the machine learning model designed to translate Sentinel-1 Synthetic Aperture Radar (SAR) based **Radar Vegetation Index (RVI)** into Sentinel-2 **Normalized Difference Vegetation Index (NDVI)**. Utilizing physical coordinates, seasonal parameters, dynamic world probabilities, and meteorological variables, the model provides cloud-free synthetic NDVI estimations to support continuous parcel-level agricultural audits.

EVALUATION METHOD
Group-based Train-Test Split (80/20)
Grouped by Parcel ID (task_id)

TRAINING COHORT
244 Train Parcels (18,289 rows)
62 Unseen Test Parcels (4,810 rows)

MODEL TYPE
Random Forest Regressor
June 2026

1. Dataset Alignment & Model Performance

PERFORMANCE SUMMARY

Due to clouds obstructing optical bands, creating a synthetic proxy of NDVI from Sentinel-1 radar bands allows for continuous agricultural health monitoring. To train the translator model, Sentinel-2 optical scenes were paired with their nearest Sentinel-1 observations within a **3-day matching window**, producing a high-quality dataset of 23,099 aligned observations.

Operational Validation: Spatial Generalization Check

By splitting training and testing data at the parcel level (Group-based Split), the model was evaluated on 62 parcels that it had never seen during training. This simulates the exact production environment where the model is requested to generate predictions for entirely new geographical boundaries.

Model Benchmarking & Comparison

Model Architecture	Partition	R^2 Score	RMSE Error	MAE Error
Random Forest Baseline (24 Features)	Train	0.9686	0.0442	0.0369
	Test	**0.8975**	**0.0919**	**0.0634**
Regularized LSTM (Standard Recurrent)	Train	0.9320	0.0651	0.0483
	Test	**0.9022**	**0.0898**	**0.0646**
Bidirectional GRU (Sequence Imputer)	Train	0.9376	0.0624	0.0465
	Test	**0.9054**	**0.0884**	**0.0632**

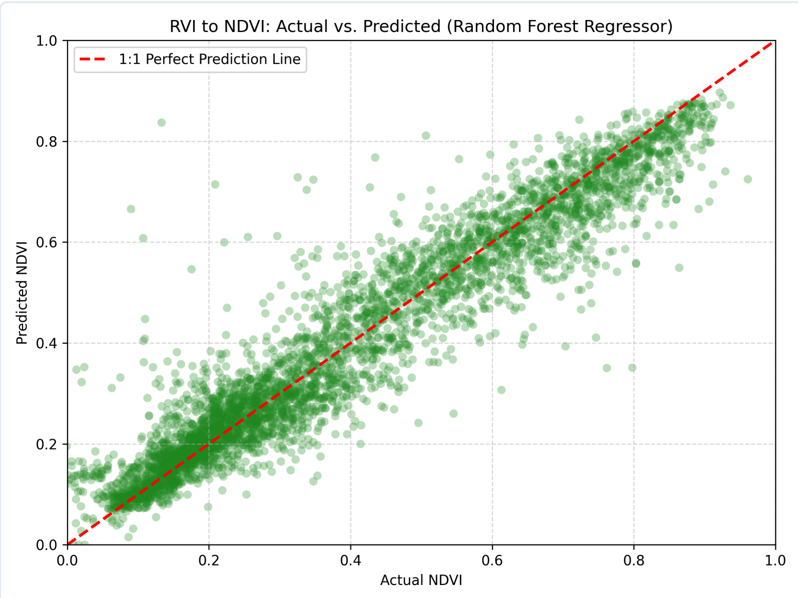

Figure 1: RVI to NDVI actual vs. predicted values on unseen test parcels.

2. Detailed Model Feature Specifications

FEATURES & DESCRIPTIONS

The RVI-to-NDVI translation model utilizes a full set of 24 spatial, structural, meteorologic, and phenological features to reconstruct vegetation indices. The table below lists all features used, what they are, and their operational role:

Feature Name	What the Feature Is	Why It Is Used in the Model
`latitude`	Absolute geographic latitude coordinate of the parcel centroid	Captures large-scale spatial/climatic gradients and regional crop calendar variations across India.
`longitude`	Absolute geographic longitude coordinate of the parcel centroid	Captures spatial gradients, local water availability, and regional agricultural boundaries.
`raw_RVI`	Sentinel-1 Radar Vegetation Index calculated from VV and VH backscatter power	Primary structural proxy of vegetation canopy density that penetrates cloud cover.
`is_kharif`	Binary seasonal flag (1 if date falls between June and October)	Represents the monsoon Kharif crop season, aligning the model with high-rainfall crop dynamics.
`is_rabi`	Binary seasonal flag (1 if date falls between November and March)	Represents the winter Rabi crop season, capturing irrigated winter crop phenology.
`is_zaid`	Binary seasonal flag (1 if date falls between April and May)	Represents the dry summer Zaid crop season, adjusting for extreme temperatures and summer crops.
`doy_sin`	Sine-transformed Day of Year (circular phenology parameter)	Captures the cyclic nature of annual crop growth, calendar progression, and solar patterns.
`doy_cos`	Cosine-transformed Day of Year (circular phenology parameter)	Provides orthogonal phase reference for DOY, ensuring continuous temporal phenology representation.
`Rainfall_15d_sum`	15-day cumulative rainfall leading up to the observation (mm)	Indicates soil moisture increases and precipitation dynamics affecting crop growth rate.
`MaxTemp_7d_avg`	7-day average of daily maximum surface temperature (°C)	Captures heat accumulation and evapotranspiration stress limits that impact canopy greenness.
`MinTemp_7d_avg`	7-day average of daily minimum surface temperature (°C)	Reflects night-time respiration rates and seasonal cooling trends affecting crop development.
`crops`	Dynamic World crop classification probability probability (0.0 to 1.0)	Provides baseline prior indicating if the parcel is active agricultural land.

2. Detailed Model Feature Specifications (Continued)

FEATURES & DESCRIPTIONS

Feature Name	What the Feature Is	Why It Is Used in the Model
`trees`	Dynamic World tree cover probability (0.0 to 1.0)	Identifies presence of perennial orchards/forests, adjusting the baseline NDVI projection.
`grass`	Dynamic World grass cover probability (0.0 to 1.0)	Distinguishes natural short-grass pasture signatures from structured crop fields.
`flooded_vegetation`	Dynamic World flooded vegetation probability (0.0 to 1.0)	Captures wetland/paddy crop dynamics where water backscatter distorts normal radar signals.
`shrub_and_scrub`	Dynamic World shrub/scrub cover probability (0.0 to 1.0)	Identifies sparse woody scrublands, distinguishing them from dense agricultural canopies.
`built`	Dynamic World built-up/infrastructure probability (0.0 to 1.0)	Detects urban or structural materials that generate high radar double-bounce reflections.
`bare`	Dynamic World bare soil probability (0.0 to 1.0)	Identifies harvested fields, fallow land, or dry soil background reflection periods.
`water`	Dynamic World open water probability (0.0 to 1.0)	Flags open water bodies, preventing specularity anomalies in VV/VH radar signals.
`snow_and_ice`	Dynamic World snow/glacier probability (0.0 to 1.0)	Prevents snow-melt or glacier signals from distorting vegetation indices in hilly regions.
`RVI_lag_12`	Linearly interpolated RVI value 12 days prior to observation date	Establishes a 12-day historical baseline of structural vegetative state to track trends.
`RVI_lag_6`	Linearly interpolated RVI value 6 days prior to observation date	Tracks short-term historical crop growth or harvest decay trajectory.
`RVI_lead_6`	Linearly interpolated RVI value 6 days ahead of observation date	Captures short-term future canopy development trends during temporal matching.
`RVI_lead_12`	Linearly interpolated RVI value 12 days ahead of observation date	Captures mid-term future canopy growth and structural senescence progression.

3. Drivers of Vegetation Reconstruction

FEATURE IMPORTANCE & LULC CONTEXT

Feature importance analysis reveals that the target's current radar value (`raw_RVI`) and its temporal dynamics (lags/leads representing recent trends) serve as the dominant indicators. Local meteorological inputs (rainfall, temperature) and dynamic land cover distributions successfully constrain regional variations in signal response.

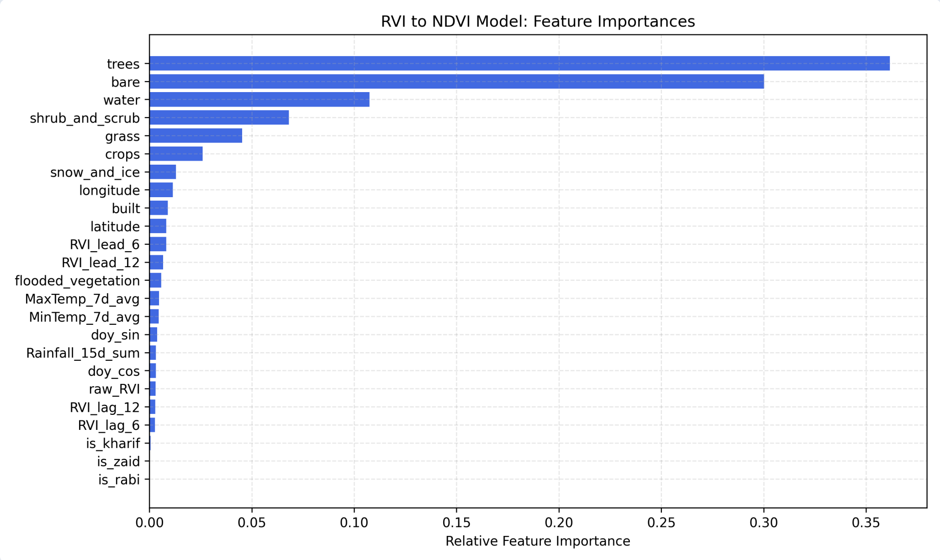

Figure 2: Relative feature importances in Random Forest regressor.

Temporal Tracking Case Study

The reconstructed NDVI curve (orange) tracks the optical NDVI signature (green) over the full course of multiple seasonal crop cycles. This highlights that the model successfully filters out sensor speckle while maintaining sensitive response times to crop emergence, peak greening, and harvest events.

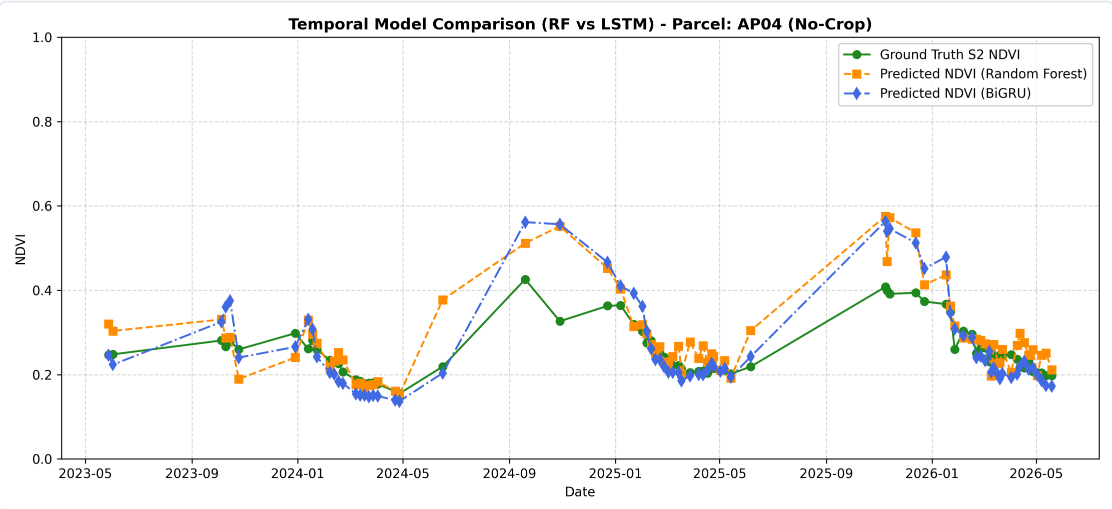

Figure 3: Temporal prediction curve comparison (Random Forest vs. BiGRU) on an unseen test parcel.

4. Deep Learning Interpretability & Diagnostics

BIGRU DIAGNOSTICS

To evaluate the deep learning model's performance and internal representation, we perform standard diagnostics on the **Bidirectional GRU (BiGRU)** network. The scatter plot below displays predicted versus actual NDVI values on the unseen test parcels. The permutation feature importance chart displays the relative influence of the 24 inputs, computed by evaluating the increase in test Mean Squared Error (MSE) when the values of each feature are shuffled.

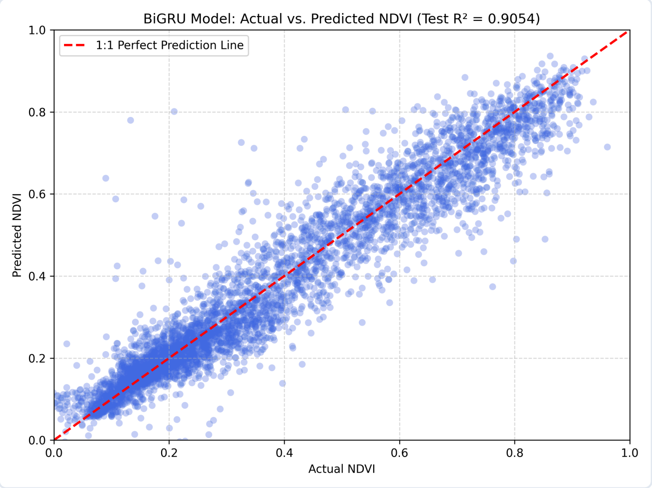

Figure 4: BiGRU Model: Actual vs. predicted NDVI values on unseen test parcels (Test R^2 = 0.9054).

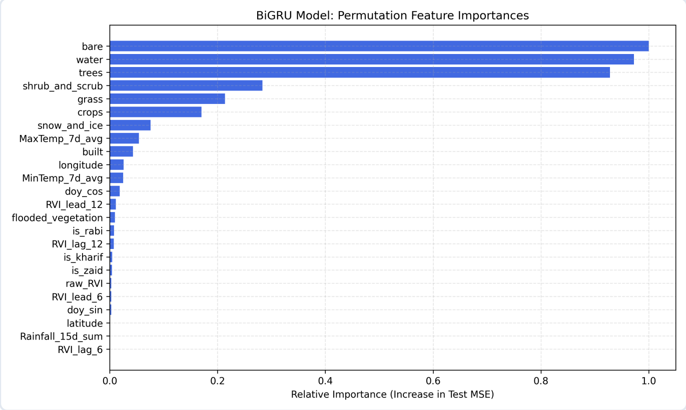

Figure 5: BiGRU Model: Permutation feature importances based on test MSE increase.

Deep Learning Interpretability

The permutation feature importance highlights that the BiGRU model relies heavily on `raw_RVI` and the forward/backward temporal context variables (`RVI_lead_6`, `RVI_lag_6`). By integrating sequential context, the model learns a more continuous representation of crop growth curves compared to decision-tree architectures, reducing sensitivity to high-frequency speckle noise and maintaining robust NDVI predictions during periods of extended cloud cover.